

Database Functions (Self-Learning)

Q. What is a User-Defined Function (UDF) in SQL? Explain how functions differ from Stored Procedures regarding parameters, return values, and execution contexts.

📌 Definition to Remember

A **User-Defined Function (UDF)** is a stored program that encapsulates SQL and procedural logic to perform a specific computation and **must return a single value** (scalar or table). Unlike procedures, functions are embedded directly inside SQL statements like `SELECT`, making them ideal for **read-only calculations and data transformation**.

1. Creating and Calling Functions

Creation: Must declare the return type explicitly.

```
CREATE FUNCTION Calculate_Bonus (salary DECIMAL)
RETURNS DECIMAL
AS
BEGIN
    RETURN salary * 0.10;  -- Must use RETURN
END;
```

Calling: Functions return a value, so they are embedded inside SQL — NOT called via `CALL`.

```
SELECT Name, Salary, Calculate_Bonus(Salary) AS Bonus FROM Employee;
```

2. Parameters and Return Values

- **Parameters:** Functions only accept **IN parameters** (input only — no OUT or INOUT).
- **Return Value:** A function **MUST** return a value using `RETURN`.
- **Scalar Function:** Returns a single value (integer, string, decimal, etc.)
- **Table-Valued Function:** Returns an entire table (supported in some RDBMS).

3. Procedures vs Functions

Feature	Stored Procedure	User-Defined Function
Return Value	Optional; can return via OUT params	Mandatory — must return a value
Called Using	EXEC or CALL	Embedded in SQL (SELECT , WHERE)
DML Operations	Can INSERT, UPDATE, DELETE	Read-only — cannot modify data
Transactions	Can use COMMIT/ROLLBACK	Cannot manage transactions
Parameters	IN, OUT, INOUT	IN only
Best For	Workflows, data modification	Calculations, formatting, read-only logic

★ Must-Write Points (for 10 marks)

1. UDF = User-Defined Function — extends SQL with custom calculations.
2. Created with **CREATE FUNCTION** ; **MUST** declare return type and use **RETURN** .
3. Functions accept only **IN parameters** (no OUT or INOUT).
4. Functions are embedded directly inside SQL statements (not called with **CALL**).
5. Functions are **read-only** — they CANNOT perform INSERT, UPDATE, or DELETE.
6. Procedures are optional return; Functions have **mandatory** return.
7. Scalar function → single value; Table-valued function → returns a table.

⚡ Quick Recall

UDF → **MUST** return value (RETURNS + RETURN) → **IN** params only → Embedded in **SELECT/WHERE** → **Read-only** (no DML) → vs Procedure: optional return, **CALL**, DML allowed